5/8/2026

Lec 3: density matrices, entanglement, Schmidt decomposition, reduced density matrices, Bell's inequality, dense coding

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(1) Density matrix

$|\psi\rangle$ – vector – pure state

Density matrix can describe both pure and mixed quantum states

$$\hat{\rho} = \sum_n p_n |\psi_n\rangle\langle\psi_n|$$

example: $\hat{\rho} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$ – mixture/mixed state

vs.

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$ – pure state

For pure states, only one p_n is nonzero

purity – describes how little (how close to zero the other 'unwanted states' are

Density matrix is not unique

$$\hat{\rho} = \frac{1}{3}|0\rangle\langle 0| + \frac{1}{3}\left(\frac{|0\rangle + \sqrt{3}|1\rangle}{2}\right)\left(\frac{\langle 0| + \sqrt{3}\langle 1|}{2}\right) + \frac{1}{3}\left(\frac{|0\rangle - \sqrt{3}|1\rangle}{2}\right)\left(\frac{\langle 0| - \sqrt{3}\langle 1|}{2}\right)$$

Trace $\quad \mathrm{Tr}\,\hat{B} = \sum_n \langle x_n | \hat{B} | x_n \rangle \quad \{|x_n\rangle\}$ orthonormal set

$$\mathrm{Tr}(\hat{\rho}\hat{A}\hat{B}) = \mathrm{Tr}(\hat{B}\hat{\rho}\hat{A}) = \mathrm{Tr}(\hat{A}\hat{B}\hat{\rho})$$

invariant under permutation

$$\mathrm{Tr}(\hat{\rho}\hat{A}\hat{B}) - \mathrm{Tr}(\hat{\rho}\hat{B}\hat{A}) = \mathrm{Tr}(\hat{\rho}[\hat{A},\hat{B}])$$

For an observable $\hat{A}$, the average value

$$\langle \hat{A} \rangle = \mathrm{Tr}(\hat{\rho}\hat{A})$$

$$= \sum_m \langle x_m | \left(\sum_n p_n |\psi_n\rangle\langle\psi_n| \right) \hat{A} | x_m \rangle$$

$$= \sum_n p_n \langle \psi_n | \hat{A} \underbrace{\sum_m |x_m\rangle\langle x_m|}_{I} |\psi_n\rangle$$

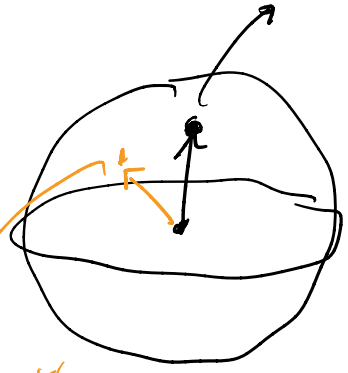
$$\langle \hat{A} \rangle = \sum_n p_n \langle \psi_n | \hat{A} | \psi_n \rangle$$

degree of purity

$\mathrm{Tr}(\hat{\rho}^2) = 1$ for pure states

< 1 for mixed states

$$\hat{\rho} = \frac{1}{2}(I + \vec{r}\cdot\vec{\sigma})$$



on the surface, the quantum state is pure

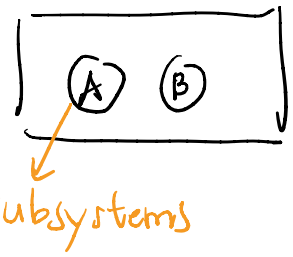
inside the Bloch sphere, the quantum state is mixed

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

Exercise: Using ↑ show that $\hat{\rho} = \frac{1}{2}\begin{bmatrix} 1+z & x-iy \\ x+iy & 1-z \end{bmatrix}$

Entanglement

$|\psi\rangle$



$|\psi\rangle = |a\rangle |b\rangle$ product state

If the state cannot be written as a product state, i.e. $|x\rangle |y\rangle$, the state is entangled.

For an operator $\hat{A}$ acting only on subsystem A

$$\hat{A} \otimes \mathbb{1}$$

$$\langle \psi | \hat{A} | \psi \rangle = \sum_m \sum_n a_m^* a_n \langle \gamma_m | \otimes \langle x_m | \hat{A} \otimes \mathbb{1} | x_n \rangle \otimes | \gamma_n \rangle$$

$$= \sum_m \sum_n a_m^* a_n \underbrace{\langle \gamma_m | \gamma_n \rangle}_{m \neq n \text{ will be } 0} \langle x_m | \hat{A} | x_n \rangle$$

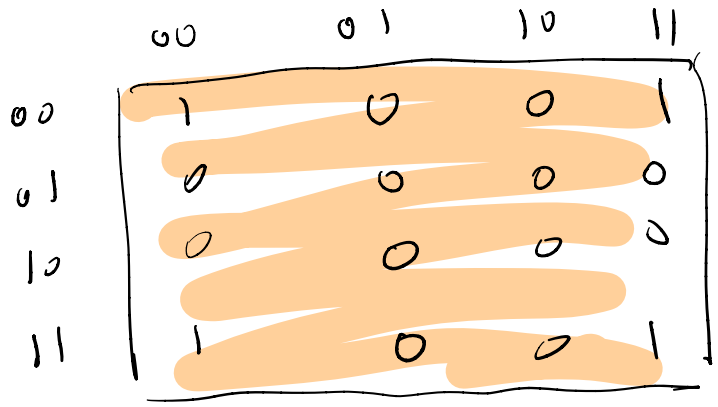
$$= \sum_n |a_n|^2 \langle x_n | \hat{A} | x_n \rangle$$

$|\psi\rangle = \sum a_n |x_n\rangle |\gamma_n\rangle$

$\langle \psi | = \sum a_m^* \langle x_m | \langle \gamma_m |$

Reduced density matrices

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) \rightarrow |\psi\rangle\langle\psi| = \frac{1}{2}\begin{cases} |00\rangle\langle 00| + |00\rangle\langle 11| \\ |11\rangle\langle 00| + |11\rangle\langle 11| \end{cases}$$



$$\rightarrow \hat{\rho}_{AB}$$

$$\hat{\rho}^A = \sum_n \langle i_n | \rho_{AB} | i_n \rangle$$

"tracing out"

$${}_B\langle 0|00\rangle\langle 00|0\rangle_B + {}_B\langle 1|00\rangle\langle 00|1\rangle_B$$

$${}_B\langle 0|00\rangle\langle 11|0\rangle_B + {}_B\langle 1|00\rangle\langle 11|1\rangle_B$$

$$+ \quad \cdots$$

$$\hat{\rho}^A = \begin{array}{c|cc} & \langle 0| & \langle 1| \\ |0\rangle & 1/2 & 0 \\ |1\rangle & 0 & 1/2 \end{array}$$

reduced density matrix

If the reduced density matrix is mixed, the quantum state of the system (A+B) is entangled.

$|\psi\rangle = \sum_n a_n |x_n\rangle |\gamma_n\rangle$ — Schmidt decomposition

I will clarify this part next workshop! The correct statement should be:

Any Schmidt decomposition with more than one nonzero coefficient will have reduced density matrices that are mixed.

basis states that ~~maximize~~ makes entanglement clearest

$\hat{\rho}^A = \sum |a_n|^2 |x_n\rangle\langle x_n|$

$\hat{\rho}^B = \sum |a_n|^2 |\gamma_n\rangle\langle \gamma_n|$

] reduced density matrices that are ~~maximally~~ mixed!

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Bell states

$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$

$|\psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

$|\psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

they are the common eigenstates of

$\sigma_x \otimes \sigma_x$

$\sigma_y \otimes \sigma_y$

$\sigma_z \otimes \sigma_z$

Next workshop: dense coding & quantum teleportation